UCL SCHOOL OF MANAGEMENT

**Module code:** MSIN0221

**Module title:** Natural Language Processing

**Assessment title:** Group Project - Proposal

**Group name:** Team 6

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## Introduction

Online customer reviews have become a central information infrastructure in the hotel and hospitality industry. These reviews help hotels to understand how consumers evaluate their own services and implicate guidance for product improvements and marketing strategy (Asma et al., 2023). In luxury contexts, where expectations are high and experiences are multifaceted, review text can be especially valuable because it captures granular judgments that are difficult to observe through internal metrics alone.

Given the importance of customer reviews, natural language processing (NLP) techniques have shown the power to extract meaningful information from the review text. Much of the NLP research on hotel reviews has been oriented toward sentiment analysis and rating-based evaluation. Dominant pipelines of NLP on hotel reviews typically map text into coarse polarity categories (e.g., positive/negative/neutral) using relatively standard preprocessing and classification methods, and uneven coverage and methodological concentration can be observed within the hospitality domain (Asma et al., 2023). As a result, low ratings or negative sentiment are often assumed to be the primary carriers of actionable critique and therefore the focus of studies.

A consequential gap follows from the fact that star ratings and textual narratives do not always align. Recent hospitality research explicitly documents score–textual inconsistency, including cases where high ratings are with comparatively low textual sentiment, suggesting that star ratings alone can be an incomplete summary of the information embedded in review text (Wang et al., 2025). Moreover, extreme high-rating reviews can be informative because they may convey a reviewer's distinctive experience and evaluative stance (Román et al., 2024). Taken together, these findings motivate closer attention to high-star reviews. Even when overall satisfaction is signaled, reviewers may still embed constructive critiques, advice, or improvement suggestions within an otherwise positive narrative.

To address this gap, this project introduces suggestion mining (SM) to the luxury hospitality domain. Defined by the SemEval-2019 shared task as the classification of sentences to identify advice and recommendations, SM remains an under-explored problem compared to sentiment analysis (Negi et al., 2019). While SemEval established benchmarks using general hotel reviews, this project specifically targets the "hidden suggestions" within high-rated reviews at luxury hotels.

The primary research question driving this project is: **To what extent can NLP models detect actionable improvement suggestions within high-star hotel reviews?** By shifting the focus from sentiment polarity to suggestion detection, this project aims to transform review data into operational actions, providing a framework for identifying improvement opportunities that are currently obscured by high satisfaction scores.

**Methodology**

The Gantt chart and task tracking table can be found in Appendix.

1. ***Data Collection and Preprocessing***
   We plan to utilize publicly available datasets (e.g. the Marina Bay Sands TripAdvisor dataset (Liu, 2022)) with over 10,000 reviews for inference. Additionally, we will also consider using an existing labeled dataset (e.g. SemEval-2019 hotel suggestion dataset) to complement our annotations later if needed. To preprocess the data, we will use the spaCy library to segment the reviews into individual sentences, as suggestion mining is strictly a sentence-level classification task. We will then filter the dataset to isolate only reviews with ratings of 4 and 5 stars, focusing our research specifically on "hidden" suggestions. Unlike standard sentiment analysis, we will strictly retain stop-words (e.g., "should," "if," "would") as these are critical linguistic indicators of the subjunctive mood often used in polite suggestions. Basic data cleaning steps such as language filtering, duplicate removal, and sentence length thresholding will also be applied.
2. ***Exploratory Data Analysis***
   We will perform statistical analysis on the linguistic features of the datasets, including the distribution of sentence lengths and the frequency of modal verbs in the labeled data compared to the unlabeled data. Visualization of these distributions will be included to understand the domain gap between the training data and our target data. We also plan to analyze the temporal distribution of the reviews to identify if review volume correlates with specific events (e.g., post-pandemic travel). Insights from EDA will inform feature selection and error analysis in subsequent modeling stages.
3. ***Annotation of Suggestions***
   We construct a labeled corpus by leveraging existing annotated datasets as references for suggestion mining. Manual annotation is conducted at the sentence level, as individual reviews may simultaneously contain praise and implicit suggestions. Following established definitions in prior work, we formulate suggestion mining as a binary sentence-level classification task (Negi et al., 2019). As a result, each sentence is assigned one of two labels: Suggestion or Not a Suggestion.
4. ***Model Development***
   Existing research emphasizes the value of evaluating neural models against rule-based baselines to better understand the contribution of contextualized representations in suggestion classification (Pecar et al., 2019). We plan to implement a BERT model for detecting suggestions automatically, which allows transfer learning with limited training data. BERT-based models have achieved strong F1 performance of 0.7812 and 0.8579 across subtasks in a similar study, indicating that pre-trained representations are effective for suggestion classification, particularly when labelled data is limited and domain transfer is required (Liu et al., 2019).
   We will also experiment with a simpler model (e.g. using Regex based on linguistic patterns) as a baseline for comparison. Once the model is trained, we will run inference on the segmented sentences to classify them as "Suggestions" or "Non-Suggestions."

5. ***Insight Generation***
   To categorize the extracted suggestions, we will apply Latent Dirichlet Allocation (LDA), which will cluster the suggestions into latent themes. We will then visualize these topics with a semantic map of the issues hidden within the positive reviews.
6. ***Evaluation and Validation***
   Model performance will be evaluated through precision, recall, and F1-Score on a held-out test set. To validate performance on actual data, we will randomly sample 100 sentences predicted as Suggestions and Non-suggestions respectively, labeling them manually, and calculate an estimated precision score. Finally, we aim to compile a short list of top suggestions identified from the positive reviews.

**Expected Outcomes**

This project is expected to deliver an NLP pipeline for identifying and analyzing implicit suggestions embedded in high rating hotel reviews. By applying sentence-level suggestion classification using BERT based model, we intend to demonstrate that positive reviews contain actionable feedback that are systematically overlooked by traditional sentiment analysis approaches. The integration of topic modeling is expected to discover recurring service themes and provide a structured overview of customer expectations. Beyond empirical findings, the project aims to provide methodological insights into the challenges of suggestion mining. Overall, this project is expected to produce a reproducible and scalable analytical framework, which highlights both the strengths and constraints of contemporary NLP techniques in customer review analysis.

**Limitations**

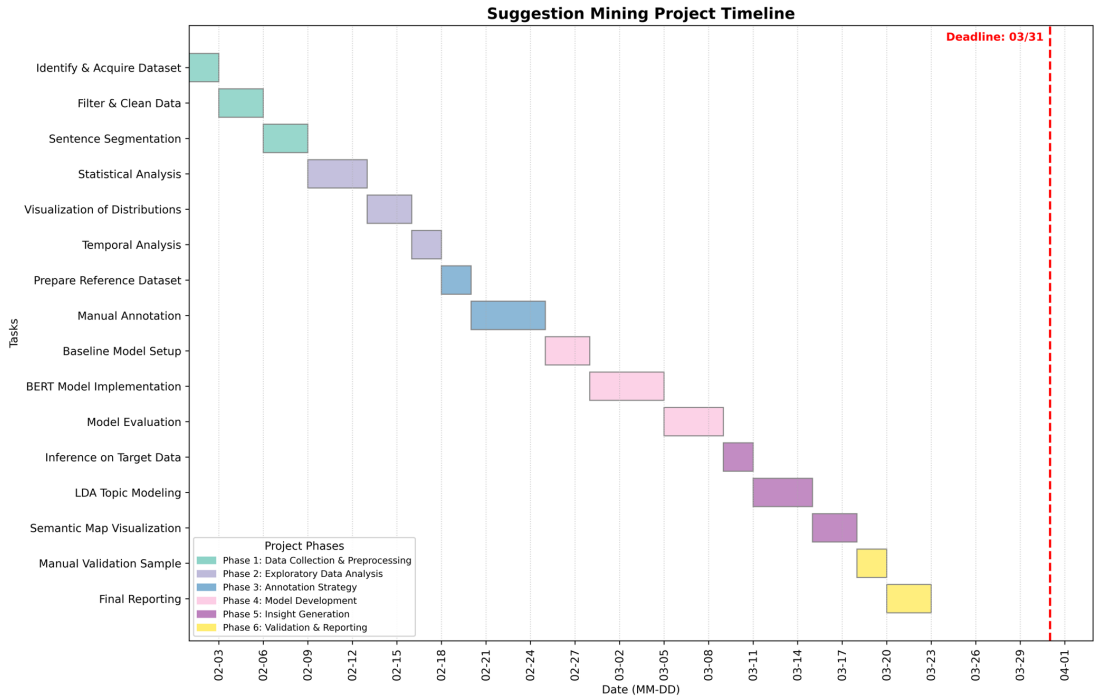
This study may face three primary challenges. First, the linguistic ambiguity of suggestion mining presents a significant hurdle. Suggestions are often expressed indirectly or politely, making it difficult to distinguish descriptive statements from genuine suggestions. While we retain functional stop words (e.g., "should," "would") to capture advisory cues, the inherent subjectivity of natural language may lead to classification errors.

Second, the pipeline architecture introduces a risk of error propagation. Misclassified sentences may also introduce noise into the subsequent topic modeling stage. It can also potentially affect the interpretability of extracted themes. This challenge is further compounded by the fact that even human annotators may disagree when interpreting implicit suggestions. Furthermore, applying LDA to short, sentence-level texts may exacerbate data sparsity issues, which may constrain semantic depth compared to document-level analysis.

Finally, the project scope involves a trade-off regarding context and generalizability. Sentence-level classification strips away broader discourse context, which can make isolated suggestions difficult to interpret. Additionally, by focusing exclusively on high-rated reviews, the findings are specific to "optimization" feedback in the luxury segment and may not generalize the explicit complaint patterns found in lower-rated reviews.

# Appendix

## Fig.1 Gantt chart




## Fig.2 Task Tracking Table

| Phase / Task Name | Duration (Days) | Dependencies | Description (Methodology Source) |
|---|---|---|---|
| **Phase 1: Data Collection & Preprocessing** | | | |
| Identify & Acquire Dataset | 2 | - | Kaggle TripAdvisor Dataset (Marina Bay Sands) |
| Filter & Clean Data | 3 | Acquire Dataset | Filter 4-5 stars; Remove duplicates |
| Sentence Segmentation | 3 | Filter Data | Use spaCy for sentence-level splitting |
| **Phase 2: Exploratory Data Analysis** | | | |
| Statistical Analysis | 4 | Segmentation | Analyze sentence length & modal verbs |
| Visualization of Distributions | 3 | Stat Analysis | Compare labeled vs. unlabeled data |
| Temporal Analysis | 2 | Visualization | Review volume vs. events (post-pandemic) |
| **Phase 3: Annotation Strategy** | | | |
| Prepare Reference Dataset | 2 | - | Load SemEval-2019 dataset |
| Manual Annotation | 5 | Reference Data | Binary labeling: Suggestion vs. Non-Suggestion |
| **Phase 4: Model Development** | | | |
| Baseline Model Setup | 3 | Annotation | Regex based on linguistic patterns |
| BERT Model Implementation | 5 | Baseline Model | Transfer learning for suggestion detection |
| Model Evaluation | 4 | BERT Model | Precision, Recall, F1-Score on test set |
| **Phase 5: Insight Generation** | | | |
| Inference on Target Data | 2 | Evaluation | Classify Marina Bay Sands sentences |
| LDA Topic Modeling | 4 | Inference | Cluster suggestions into latent themes |
| Semantic Map Visualization | 3 | LDA Model | Visualizing issues in positive reviews |
| **Phase 6: Validation & Reporting** | | | |
| Manual Validation Sample | 2 | Semantic Map | Validate 100 random predictions |
| Final Reporting | 3 | Validation | Compile top suggestions list |

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